
Rufin VANRULLEN

Peer-reviewed Articles since 2016

1. Butlin, P., Bayne, T., Fleming, S. M., Mudrik, L., Peters, M. A., Schwitzgebel, E., ... & VanRullen, R. (2026). Consciousness indicators, mimicry, and internal variants. *Trends Cogn. Sci.* (in press)
2. VanRullen, R. (2026). AI Consciousness and Existential Risk. *Journal of AI Consciousness* (in press).
3. Bertin-Johannet, R., Scipio, L., Maytié, L., & VanRullen, R. (2026). An Attention Mechanism for Robust Multimodal Integration in a Global Workspace Architecture. *Lecture Notes in Computer Science* (in press).
4. Butlin, P., Long, R., Bayne, T., Bengio, Y., Birch, J., Chalmers, D., ... & VanRullen, R. (2026). Identifying indicators of consciousness in AI systems. *Trends Cogn. Sci.* 30(6), 488-501
5. Béthune, L., Vigouroux, D., Du, Y., VanRullen, R., Serre, T., & Boutin, V. (2025). Follow the Energy, Find the Path: Riemannian Metrics from Energy-Based Models. *NeurIPS 2025: Advances in Neural Information Processing Systems*, 38.
6. Ferrante, M., Boccatto, T., Toschi, N. & VanRullen, R. (2025). Evidence for compositionality in fMRI visual representations via Brain Algebra. *Communications Biology* 8(1), 1263.
7. Nikolaus, M., Mozafari, M., Berry, I., Asher, N., Reddy, L., & VanRullen, R. (2025). Modality-Agnostic Decoding of Vision and Language from fMRI. *eLife*, 14.
8. Muzellec, S., Alamia, A., Serre, T., & VanRullen, R. (2025). Enhancing deep neural networks through complex-valued representations and Kuramoto synchronization dynamics. *TMLR: Transactions in Machine Learning Research*
9. Muzellec, S., Linsley, D., Ashok, A. K., Mingolla, E., Malik, G., VanRullen, R., & Serre, T. (2025). Tracking objects that change in appearance with phase synchrony. *ICLR 2025: International Conference on Learning Representations*. arXiv:2410.02094.
10. Alamia, A., Dalliès, L., Faye, G., & VanRullen, R. (2025). Wave Propagation Phenomena in Nonlinear Hierarchical Neural Networks with Predictive Coding Feedback Dynamics. *Bull Mathematical Biology*, 87(9), 115.
11. Boutin, V., Mukherji, R., Agrawal, A., Muzellec, S., Fel, T., Serre, T., & VanRullen, R. (2024). Latent Representation Matters: Human-like Sketches in One-shot Drawing Tasks. *NeurIPS 2024: Advances in Neural Information Processing Systems*, 37, 96282-96324.
12. Devillers, B., Maytié, L., & VanRullen, R. (2024). Semi-supervised multimodal representation learning through a global workspace. *IEEE Transactions on Neural Networks and Learning Systems*. 36 (5), 7843-7857.doi: 10.1109/TNNLS.2024.3416701 (in press)
13. Maytié, L., Devillers, B., Arnold, A., & VanRullen, R. (2024). Zero-shot cross-modal transfer of Reinforcement Learning policies through a Global Workspace. *RLC 2024: Reinforcement Learning Journal*, 3(170), 1410-1426.*Conference*. arXiv:2403.04588.
14. Muzellec, S., Andéol, L., Fel, T., VanRullen, R., & Serre, T. (2024). Saliency strikes back: How filtering out high frequencies improves explanations. *ICML 2024: International Conference on Machine Learning*, arXiv:2307.09591.
15. Ferrante, M., Boccatto, T., Ozcelik, F., VanRullen, R., & Toschi, N. (2024). Through their eyes: multi-subject Brain Decoding with simple alignment techniques. *Imaging Neuroscience*, 2, 1-21.
16. Decourt, C., VanRullen, R., Salle, D., & Oberlin, T. (2024). A recurrent CNN for online object detection on raw radar frames. *IEEE Transactions on Intelligent Transportation Systems*. 25(10), 13432-13441. doi: 10.1109/TITS.2024.3404076 (in press).
17. Alamia, A., & VanRullen, R. (2024). A Traveling Waves Perspective on Temporal Binding. *J Cogn Neurosci*, 36(4), 721-729.
18. Chalvidal, M., Serre, T., & VanRullen, R. (2024). Learning functional transduction. *NeurIPS 2023: Advances in Neural Information Processing Systems*, 3637.
19. Faye, G., Foulhé, G., & VanRullen, R. (2023). Mathematical derivation of wave propagation properties in hierarchical neural networks with predictive coding feedback dynamics. *Bull Mathematical Biology*, 85:80
20. Ozcelik, F., & VanRullen, R. (2023). Natural scene reconstruction from fMRI signals using generative latent diffusion. *Scientific Reports*, 13(1), 15666.
21. Bielawski, R., & Vanrullen, R. (2023). CLIP-based image captioning via unsupervised cycle-consistency in the latent space. ACL 2023: 8th Workshop on Representation Learning for NLP (*RepL4NLP 2023*), 266-275.
22. Alamia, A., Terral, L., D'ambra, M. R., & VanRullen, R. (2023). Distinct roles of forward and backward alpha-band waves in spatial visual attention. *Elife*, 12, e85035.
23. Chota, S., VanRullen, R.* & Gulbinaite, R.* (2023). Random tactile noise stimulation reveals beta-rhythmic impulse response function of the somatosensory system. *J Neurosci*, 43(17), 3107-3119 .
24. Alamia, A., Mozafari, M., Choksi, B., & VanRullen, R. (2022). On the role of feedback in image recognition under noise and adversarial attacks: A predictive coding perspective. *Neural Networks*. doi: 10.1016/j.neunet.2022.10.020
25. Chalvidal, M., Serre, T., & VanRullen, R. (2022). Meta-Reinforcement Learning with Self-Modifying Networks. *NeurIPS 2022: Advances in Neural Information Processing Systems*, arXiv:2202.02363.
26. Choksi, B., Mozafari, M., VanRullen, R., & Reddy, L. (2022). Multimodal neural networks better explain multivoxel patterns in the hippocampus. *Neural Networks*. doi: 10.1016/j.neunet.2022.07.033
27. Ozcelik, F., Choksi, B., Mozafari, M., Reddy, L., & VanRullen, R. (2022). Reconstruction of perceived images from fMRI patterns and semantic brain exploration using Instance-Conditioned GANs. *Int. Joint Conf. Neural Networks (IJCNN)*.
28. VanRullen, R. (2022). Deep learning in alternate reality. *Nature Human Behaviour*, 6(1), 27-28.
29. Vaishnav, M., Cadene, R., Alamia, A., Linsley, D., VanRullen, R., & Serre, T. (2022). Understanding the computational demands underlying visual reasoning. *Neural Computation*, 34 (5), 1075-1099
30. Merholz, G., Grabot, L., VanRullen, R., & Dugué, L. (2022). Periodic attention operates faster during more complex visual search. *Scientific Reports*, 12(1), 1-14.
31. Bielawski, R., Devillers, B., Van de Cruys, T., & VanRullen, R. (2022). When does CLIP generalize better than unimodal models? When judging human-centric concepts. ACL 2022: 7th workshop on Representation Learning for NLP (*Rep4NLP 2022*), 29-38

32. Decourt, C., VanRullen, R., Salle, D., & Oberlin, T. (2022). DAROD: a deep automotive radar object detector on range-doppler maps. *Proceedings of the 33rd IEEE Intelligent Vehicles Symposium (IV 2022)*, 112-118.
33. Fakche, C., VanRullen, R., Marque, P., & Dugué, L. (2022). Alpha Phase-Amplitude Tradeoffs Predict Visual Perception. *eNeuro*, 9(1).
34. Luo, C., Chen, W., VanRullen, R., Zhang, Y., & Gaspar, C. M. (2022). Nudging the N170 forward with prior stimulation—Bridging the gap between N170 and recognition potential. *Human brain mapping*, 43(4), 1214-1230.
35. Choksi, B., Mozafari, M., Biggs O'May, C., Ador, B., Alamia, A., & VanRullen, R. (2021). Predify: Augmenting deep neural networks with brain-inspired predictive coding dynamics. *NeurIPS 2021: Advances in Neural Information Processing Systems*, 34.
36. Pang, Z., Biggs O'May, C., Choksi, B. & VanRullen, R. (2021). Predictive coding feedback results in perceived illusory contours in a recurrent neural network. *Neural Networks*. doi: 10.1016/j.neunet.2021.08.024.
37. VanRullen, R., & Kanai, R. (2021). Deep learning and the Global Workspace Theory. *Trends in Neurosciences*. 44(9), 692-704
38. VanRullen, R., & Alamia, A. (2021). GAttANet: Global attention agreement for convolutional neural networks. *Lecture Notes in Computer Science: Artificial Neural Networks and Machine Learning*, 12891, 1-11.
39. Choksi, B., Mozafari, M., VanRullen, R., & Reddy, L. (2021). Multimodal neural networks better explain multivoxel patterns in the hippocampus. In Neural Information Processing Systems (NeurIPS) conference: 3rd Workshop on *Shared Visual Representations in Human and Machine Intelligence* (SVRHM 2021).
40. Reddy, L., Self, M.W., Zoefel, B., Poncet, M., Possel, J.K., Peters, J.C., Baayen, J.C., Idema, S., VanRullen, R. and Roelfsema, P.R., (2021). Theta-phase dependent neuronal coding during sequence learning in human single neurons. *Nature Communications*, 12(1), 1-9.
41. Devillers, B., Choksi, B., Bielawski, R., & VanRullen, R. (2021). Does language help generalization in vision models? *CoNLL2021: Proceedings of the 25th Conference on Computational Natural Language Learning*, 171-182.
42. Reddy, L., Cichy, R. M., & VanRullen, R. (2021). Representational content of oscillatory brain activity during object recognition: contrasting cortical and deep neural network hierarchies. *eNeuro*, 8(3).
43. Chalvidal, M., Ricci, M., VanRullen, R., & Serre, T. (2021). Go with the flow: Adaptive control for Neural ODEs. *ICLR 2021: International Conference on Learning Representations*.
44. Luo, C., Brüers, S., Berry, I., VanRullen, R., & Reddy, L. (2021). Tentative fMRI signatures of perceptual echoes in early visual cortex. *NeuroImage*, 237, 118053.
45. Luo, C., VanRullen, R., & Alamia, A. (2021). Conscious perception and perceptual echoes: a binocular rivalry study. *Neuroscience of Consciousness*, 2021(1), niab007.
46. Chota, S., Marque, P., & VanRullen, R. (2021). Occipital Alpha-TMS causally modulates Temporal Order Judgements: Evidence for discrete temporal windows in vision. *NeuroImage*, 237, 118173.
47. Alamia, A., Luo, C., Ricci, M., Kim, J., Serre, T., & VanRullen, R. (2021). Differential involvement of EEG oscillatory components in sameness vs. spatial-relation visual reasoning tasks. *eNeuro*, 8(1).
48. Alamia, A., Timmermann, C., Nutt, D. J., VanRullen, R.*, & Carhart-Harris, R. L.* (2020). DMT alters cortical travelling waves. *eLife* 9, e64623.
49. Pang, Z., Alamia, A., & VanRullen, R. (2020). Turning the Stimulus On and Off Changes the Direction of α Traveling Waves. *eNeuro* 7(6).
50. Alamia, A., Gauducheu, V., Paisios, D., & VanRullen, R. (2020). Comparing feedforward and recurrent neural network architectures with human behavior in artificial grammar learning. *Scientific Reports* 10(1), 1-15.
51. Schwenk, J. C., VanRullen, R., & Bremmer, F. (2020). Dynamics of visual perceptual echoes following short-term visual deprivation. *Cerebral Cortex Communications*.
52. Mozafari, M., Reddy, L., & VanRullen, R. (2020). Reconstructing Natural Scenes from fMRI Patterns using BigBiGAN. *Int. Joint Conf. Neural Networks (IJCNN)*.
53. Chota, S., McLelland, D., Lavergne, L., Zimmermann, E., Cavanagh, P., & VanRullen, R. (2020). Full Field Masking Causes Reversals in Perceived Event Order. *Front. Neurosci.*, 14, 217.
54. Gaillard, C., Hassen, S. B. H., Di Bello, F., Bihan-Poudec, Y., VanRullen, R., & Hamed, S. B. (2020). Prefrontal attentional saccades explore space rhythmically. *Nature Comm*, 11(1), 1-13.
55. Alamia, A., & VanRullen, R. (2019). Alpha oscillations and traveling waves: Signatures of predictive coding? *PLoS Biol.*, 17(10).
56. VanRullen, R., Reddy, L. (2019). Reconstructing faces from fMRI patterns using deep generative neural networks *Communications Biology* 2(1), 193.
57. Gulbinaite, R., Roozendaal, D. H. M., & VanRullen, R. (2019). Attention differentially modulates the amplitude of resonance frequencies in the visual cortex. *Neuroimage*, 203(116146).
58. Alamia, A., VanRullen, R., Paqualato, E., Mouraux, A., Zenon, A. (2019). Pupil-linked arousal responds to unconscious surprisal. *J Neurosci*, 39(27), 5369-5376.
59. Dugué, L., Beck, A., Marque, P. & VanRullen, R. (2019). Contribution of FEF to attentional periodicity during visual search: a TMS study. *eNeuro*, 6(3).
60. Chota, S. & VanRullen, R. (2019). Visual entrainment at 10 Hz causes periodic modulation of the flash lag illusion. *Front. Neurosci.* 13(232).
61. Lozano-Soldevilla, D. & VanRullen, R. (2019). The hidden spatial dimension of alpha: 10 Hz perceptual echoes propagate as periodic travelling waves in the human brain. *Cell Reports*, 26(2), 374-380.
62. Alamia, A., Zenon, A., VanRullen, R., Duque, J. & Derosiere, G. (2018). Implicit visual cues tune oscillatory motor activity during decision-making. *Neuroimage*, 186, 424-436.
63. Chota, S., Luo, C., Crouzet, S.M., Boyer, L., Kienitz, R., Schmid, M.C. & VanRullen, R. (2018). Rhythmic fluctuations of saccadic reaction time arising from visual competition. *Scientific Reports* 8(1), 15889
64. VanRullen, R. (2018). Attention cycles. *Neuron*, 99(4), 632-634.
65. Bruers, S. & VanRullen, R. (2018). Alpha Power Modulates Perception Independently of Endogenous Factors. *Front. Neurosci.* 12:279.
66. Baures, R., Balestra, M., Rosito, M. & VanRullen, R. (2018). The detrimental influence of attention on time-to-contact perception. *Att. Percept. Psychophys.* 80(6), 1591-1598.

67. **Benedetto, A., Lozano-Soldevilla, D. & VanRullen, R. (2018).** Different responses of spontaneous and stimulus-related alpha activity to ambient luminance changes. *Eur J Neurosci*, 48(7), 2599-2608.
68. **Edwards, G., VanRullen, R. & Cavanagh, P. (2018).** Decoding trans-saccadic memory. *J Neurosci* 38(5), 1114-1123.
69. **Gulbinaite, R., van Viegen, T., Wieling, M., Cohen, M.X. & VanRullen, R. (2017).** Individual alpha peak frequency predicts 10 Hz flicker effects on selective attention. *J Neurosci*, 37(42) 10173-10184
70. **Gulbinaite, R., Ilhan, B. & VanRullen, R. (2017).** The triple-flash illusion reveals a driving role of alpha-band reverberations in visual perception. *J Neurosci* 37(30), 7219-7230.
71. **Chang, A., Schwartzman, D.J., VanRullen, R., Kanai, R. & Seth, A.K. (2017).** Visual perceptual echo reflects learning of regularities in rapid luminance sequences. *J Neurosci* 37(35), 8486-8497.
72. **Brüers, S. & VanRullen, R. (2017).** At what latency does the phase of brain oscillations influence perception? *eNeuro* 4(3):e0078-17.2017.
73. **Dugué, L. & VanRullen, R. (2017).** Transcranial Magnetic Stimulation reveals intrinsic perceptual and attentional rhythms. *Front. Neurosci.* 11:154.
74. **Zoefel, B. & VanRullen, R. (2017).** Oscillatory mechanisms of stimulus processing and selection in the visual and auditory systems: State-of-the-art, speculations and suggestions. *Front. Neurosci.* 11:296.
75. **Edwards, G., Paeye, C., Marque, P., VanRullen, R. & Cavanagh, P. (2017).** Predictive position computations mediated by parietal areas: TMS evidence. *Neuroimage* 153, 49-57
76. **VanRullen, R. (2017).** Perception Science in the age of Deep Neural Networks. *Front. Psychology* 8:142.
77. **Zoefel, B., Costa-Faidella, J., Lakatos, P., Schroeder, C.E. & VanRullen, R. (2017).** Characterization of neural entrainment to speech with and without slow spectral energy fluctuations in laminar recordings in monkey A1. *NeuroImage* 150, 344-357.
78. **Han, B. & VanRullen, R. (2017).** The rhythms of predictive coding? Pre-stimulus phase modulates the influence of shape perception on luminance judgments. *Scientific Reports* 7, 43573.
79. **VanRullen, R. (2016).** Perceptual cycles. *Trends Cogn Sci* 20(10), 723-735.
80. **McLelland, D. & VanRullen, R. (2016).** Theta-Gamma Coding Meets Communication-through-Coherence: Neuronal Oscillatory Multiplexing Theories Reconciled. *PLoS Comput Biol* 12(10), e1005162
81. **Sun, H-M., Inyutina, M., VanRullen, R. & Wu, C-T. (2016).** The temporal advantage for reloading vs. uploading conscious representations decays over time. *Neurosci Consc* 216(1), niw017.
82. **VanRullen, R. (2016).** How to evaluate phase differences between trial groups in ongoing electrophysiological signals. *Front. Neurosci* 10:426.
83. **Sherman, M. T., Kanai, R., Seth, A. K., & VanRullen, R. (2016).** Rhythmic influence of top-down perceptual priors in the phase of pre-stimulus occipital alpha oscillations. *J Cogn Neurosci*, 28(9), 1318-1330.
84. **McLelland, D., Lavergne, L. & VanRullen, R. (2016).** The phase of ongoing EEG oscillations predicts the amplitude of peri-saccadic mislocalization. *Scientific Reports* 6, 29335.
85. **Sokoliuk, R. & VanRullen, R. (2016).** Global and local oscillatory entrainment of visual behavior across retinotopic space. *Scientific Reports* 6, 25132.
86. **Han, B. & VanRullen, R. (2016).** Shape perception enhances perceived contrast: evidence for excitatory predictive feedback? *Scientific Reports* 6, 22944.
87. **Miconi, T. & VanRullen, R. (2016).** A Feedback Model of Attention Explains the Diverse Effects of Attention on Neural Firing Rates and Receptive Field Structure. *PLoS Comput Biol* 12(2): e1004770.
88. **Senoussi, M., Berry, I., VanRullen, R. & Reddy, L. (2016).** Multivoxel Object Representations in Adult Human Visual Cortex Are Flexible: An Associative Learning Study, *J Cogn Neurosci* 28(6), 852-868.
89. **Zoefel, B. & VanRullen, R. (2016).** EEG oscillations entrain their phase to high-level features of speech sound. *Neuroimage*, 124, 16-13.

Book Chapters since 2016

1. **Kuske, N., & VanRullen, R. (2026).** Consciousness in Artificial Systems: Bridging Global Workspace and Sensorimotor Theory in In-Silico Models. In *Philosophy of Artificial Intelligence: The State of the Art* (pp. 321-336). Cham: Springer Nature Switzerland.
2. **Sokoliuk, R. & VanRullen, R. (2019).** Perceptual illusions caused by discrete sampling. In *"The illusions of Time: Philosophical and Psychological Essays on Timing and Time Perception"*, Eds: A. Bardou, V. Arstila, S. Power & A. Vatakis. Palgrave Macmillan, pp. 315-338.
3. **VanRullen, R. (2018).** Perceptual Rhythms, in *"Stevens Handbook of Experimental Psychology and Cognitive Neuroscience, Vol 2: Sensation, Perception and Attention"* Ed: J. Serences. Wiley, doi:10.1002/9781119170174.epcn212.

Invited Oral Presentations in International Conferences since 2016

1. **VanRullen, R. (2026).** The Global Latent Workspace: a model of cognition with AI applications. Invited presentation, GDR SoC2 annual meeting (Marseille, France).
2. **VanRullen, R. (2026).** The Global Latent Workspace: a model of cognition with AI applications. Invited presentation, 1st Workshop on Computational Psychology (ENS Paris Saclay, France).
3. **VanRullen, R. (2025).** The Global Latent Workspace: a model of cognition with AI applications. Invited presentation, MAIN 2025 (Montreal AI and Neuroscience conference, Canada).
4. **VanRullen, R. (2025).** The Global Latent Workspace: a model of cognition with AI applications. Invited presentation, Mini-Symposium on Cognition and Computation: Bidirectional Insights Between Cognitive Neuroscience and AI. Fields Institute, Toronto (Canada).
5. **VanRullen, R. (2025).** The Global Latent Workspace: a model of cognition with AI applications. Invited presentation, AI and Consciousness Workshop (Paris, France).
6. **VanRullen, R. (2025).** Brain decoding with Generative AI. Invited presentation, INS2I seminar on Generative AI (Paris, France).
7. **VanRullen, R. (2024).** Brain-inspired multimodal deep learning. Invited presentation, Brain, Language and AI workshop, Xishuangbanna (China).

8. **VanRullen, R. (2024).** Multimodal AI systems with a Global Workspace. Invited presentation, eBrains Workshop (Marseille, France).
9. **VanRullen, R. (2024).** Deep learning and the Global Workspace Theory. Invited presentation, AISSAI workshop on Heterogeneous Data and Large Representation Models in Science, Toulouse (France).
10. **VanRullen, R. (2024).** Multimodal deep learning through a global workspace. Invited presentation, Neuro-AI: Bridging the gap between human and machine Intelligence (Amsterdam, Netherlands).
11. **VanRullen, R. (2024).** Multimodal deep learning through a global workspace. Invited presentation, Neuroscience & Artificial Intelligence (Bordeaux, France).
12. **VanRullen, R. (2024).** Brain-inspired multimodal deep learning. Invited presentation, Modeling of Life: from the atom to the animal (Toulouse, France).
13. **VanRullen, R. (2023).** Deep learning and bio-inspired computation. Invited presentation, BioComp 2023 (Banyuls, France).
14. **VanRullen, R. (2022).** Deep predictive coding for more robust and human-like vision, Invited presentation, IS 2022: SIAM conference on Imaging Science (online).
15. **VanRullen, R. (2021).** Do deep learning latent spaces resemble human brain representations? NeuroCog 2021 (Louvain, Belgium).
16. **VanRullen, R. (2021).** Deep predictive coding for more robust and human-like vision, Invited presentation, SMB 2021: Annual meeting of the Society for Mathematical Biology (online).
17. **VanRullen, R. (2019).** Perceptual Cycles, Waves and Predictive Coding, Invited presentation, Asia-Pacific Conference on Vision (Osaka, Japan).
18. **VanRullen, R. (2019).** Perceptual Cycles, Waves and Predictive Coding, Invited presentation, York University VISTA - Centre for Vision Research International Conference on Predictive Vision, Toronto (Canada).
19. **VanRullen, R. (2019).** Perceptual Cycles, Waves and Predictive Coding, Invited presentation, Dynamics of Vision and Touch (DyViTo), Rauischholzhausen (Germany).
20. **VanRullen, R. (2018).** Perceptual Cycles, Waves and Predictive Coding, Invited presentation, Alpha Scales workshop, European Institute of Theoretical Neuroscience, Paris (France).
21. **VanRullen, R. (2018).** Perceptual Cycles, Waves and Predictive Coding, Invited presentation, Symposium on New trends in decision-making: decision as inference? Paris (France).
22. **VanRullen, R. (2017).** Perceptual Cycles and Waves, Invited presentation, Timing Research Forum, Strasbourg (France).
23. **VanRullen, R. (2017).** Perceptual Cycles, Invited keynote presentation, Association for the Scientific Study of Consciousness meeting, Beijing, (China).
24. **VanRullen, R. (2017).** Perceptual Cycles in Vision and Audition, Invited presentation, Neural Oscillations in Speech and Language Processing Symposium, Berlin (Germany).

Conference Proceedings since 2016:

1. **Bertin-Johannet, R., Scipio, L., Maytié, L., & VanRullen, R. (2026).** An Attention Mechanism for Robust Multimodal Integration in a Global Workspace Architecture. ICANN 2026: International Conference on Artificial Neural Networks.
2. **VanRullen, R. (2026).** The Global Latent Workspace: a model of cognition with AI applications. Invited presentation, 1st Workshop on Computational Psychology (ENS Paris Saclay, France).
3. **Béthune, L., Vigouroux, D., Du, Y., VanRullen, R., Serre, T., & Boutin, V. (2025).** Follow the Energy, Find the Path: Riemannian Metrics from Energy-Based Models. *NeurIPS 2025: Advances in Neural Information Processing Systems*, 38.
4. **VanRullen, R. (2025).** The Global Latent Workspace: a model of cognition with AI applications. Invited presentation, MAIN 2025 (Montreal AI and Neuroscience conference, Canada).
5. **VanRullen, R. (2025).** The Global Latent Workspace: a model of cognition with AI applications. Invited presentation, Mini-Symposium on Cognition and Computation: Bidirectional Insights Between Cognitive Neuroscience and AI. Fields Institute, Toronto (Canada).
6. **Kuske, N. & VanRullen, R. (2025).** Sensorimotor Affordances in a Global Latent Workspace. Cognitive Computational Neuroscience (CCN 2025), Amsterdam (Netherlands).
7. **Nikolaus, M., Mozafari, M., Asher, N., Reddy, L. & VanRullen, R. (2025).** Modality-Agnostic Representations are Widespread Across the Cortex. Cognitive Computational Neuroscience (CCN 2025), Amsterdam (Netherlands).
8. **Kong, Y., Petras, K., Marque, P., VanRullen, R. & Dugué, L. (2025).** Oscillatory Traveling Waves in Visual Search. Vision Sciences Society (VSS) Annual Meeting, St Pete Beach (USA)
9. **VanRullen, R. (2025).** The Global Latent Workspace: a model of cognition with AI applications. Invited presentation, AI and Consciousness Workshop (Paris, France).
10. **Muzellec, S., Linsley, D., Ashok, A. K., Mingolla, E., Malik, G., VanRullen, R., & Serre, T. (2025).** Tracking objects that change in appearance with phase synchrony. *ICLR 2025: International Conference on Learning Representations*. arXiv:2410.02094.
11. **VanRullen, R. (2025).** Brain decoding with Generative AI. Invited presentation, INS2I seminar on Generative AI (Paris, France).
12. **VanRullen, R. (2024).** Brain-inspired multimodal deep learning. Invited presentation, Brain, Language and AI workshop, Xishuangbanna (China).
13. **VanRullen, R. (2024).** Multimodal AI systems with a Global Workspace. Invited presentation, eBrains Workshop (Marseille, France).
14. **Boutin, V., Mukherji, R., Agrawal, A., Muzellec, S., Fel, T., Serre, T., & VanRullen, R. (2024).** Latent Representation Matters: Human-like Sketches in One-shot Drawing Tasks. *NeurIPS 2024: Advances in Neural Information Processing Systems*, 37, 96282-96324.
15. **VanRullen, R. (2024).** Deep learning and the Global Workspace Theory. Invited presentation, AISSAI workshop on Heterogeneous Data and Large Representation Models in Science, Toulouse (France).
16. **Maytié, L., Devillers, B., Arnold, A., & VanRullen, R. (2024).** Zero-shot cross-modal transfer of Reinforcement Learning policies through a Global Workspace. *RLC 2024: Reinforcement Learning Conference*. arXiv:2403.04588.
17. **Muzellec, S., Andéol, L., Fel, T., VanRullen, R., & Serre, T. (2024).** Saliency strikes back: How filtering out high frequencies improves explanations. *ICML 2024: International Conference on Machine Learning*, arXiv:2307.09591.
18. **VanRullen, R. (2024).** Multimodal deep learning through a global workspace. Invited presentation, Neuro-AI: Bridging the gap between human and machine Intelligence (Amsterdam, Netherlands).
19. **VanRullen, R. (2024).** Multimodal deep learning through a global workspace. Invited presentation, Neuroscience & Artificial Intelligence (Bordeaux, France).
20. **VanRullen, R. (2024).** Brain-inspired multimodal deep learning. Invited presentation, Modeling of Life: from the atom to the animal (Toulouse, France).
21. **Chalvidal, M., Serre, T., & VanRullen, R. (2024).** Learning functional transduction. *NeurIPS 2023: Advances in Neural Information Processing Systems*, 37.
22. **VanRullen, R. (2023).** Deep learning and bio-inspired computation. Invited presentation, BioComp 2023 (Banyuls, France).

23. **Muzellec, S., Alamia, A., Serre, T. & VanRullen, R. (2023).** Benefits of synchrony: Improving deep neural networks using complex values and Kuramoto synchronization. *Cognitive Computational Neuroscience (CCN 2023)*, Oxford (UK).
24. **Bielawski, R., & Vanrullen, R. (2023).** CLIP-based image captioning via unsupervised cycle-consistency in the latent space. *ACL 2023: 8th Workshop on Representation Learning for NLP (RepL4NLP 2023)*.
25. **Chalvidal, M., Serre, T., & VanRullen, R. (2022).** Meta-Reinforcement Learning with Self-Modifying Networks. *NeurIPS 2022: Advances in Neural Information Processing Systems*, 35.
26. **Alamia, A., Mozafari, M., Choksi, B. & VanRullen, R. (2022).** On the role of feedback in visual processing: a predictive coding perspective. *Cognitive Computational Neuroscience (CCN 2022)*, San Francisco (USA).
27. **Choksi, B., VanRullen, R. & Reddy, L. (2022).** Do multimodal neural networks better explain human visual representations than vision-only networks? *CCN Cognitive Computational Neuroscience (CCN 2022)*, San Francisco (USA).
28. **Muzellec, S., Chalvidal, M., Serre, T. & VanRullen, R. (2022).** Accurate implementation of computational neuroscience models through neural ODEs. *Cognitive Computational Neuroscience (CCN 2022)*, San Francisco (USA).
29. **Ozcelik, F., Choksi, B., Mozafari, M., Reddy, L., & VanRullen, R. (2022).** Reconstruction of perceived images from fMRI patterns and semantic brain exploration using Instance-Conditioned GANs. *International Joint Conference on Neural Networks (IJCNN)*.
30. **Bielawski, R., Devillers, B., Van de Cruys, T., & VanRullen, R. (2022).** When does CLIP generalize better than unimodal models? When judging human-centric concepts. *ACL 2022: 5th workshop on Representation Learning for NLP (Rep4NLP 2022)*
31. **Decourt, C., VanRullen, R., Salle, D., & Oberlin, T. (2022).** DAROD: a deep automotive radar object detector on range-doppler maps. *Proceedings of the 33rd IEEE Intelligent Vehicles Symposium (IV 2022)*.
32. **VanRullen, R. (2022).** Deep predictive coding for more robust and human-like vision, Invited presentation, IS 2022: SIAM conference on Imaging Science (online).
33. **VanRullen, R. (2021).** Do deep learning latent spaces resemble human brain representations? *NeuroCog 2021 (Louvain, Belgium)*.
34. **VanRullen, R. (2021).** Deep predictive coding for more robust and human-like vision, Invited presentation, SMB 2021: Annual meeting of the Society for Mathematical Biology (online).
35. **Devillers, B., Choksi, B., Bielawski, R., & VanRullen, R. (2021).** Does language help generalization in vision models? *CoNLL2021: Proceedings of the 25th Conference on Computational Natural Language Learning*, 171-182.
36. **VanRullen, R., & Alamia, A. (2021).** GAtANet: Global attention agreement for convolutional neural networks. *ICANN 2021: International Conference on Artificial Neural Networks*.
37. **Choksi, B., Mozafari, M., Biggs O'May, C., Ador, B., Alamia, A., & VanRullen, R. (2021).** Predify: Augmenting deep neural networks with brain-inspired predictive coding dynamics. *NeurIPS 2021: Advances in Neural Information Processing Systems*, 34.
38. **Devillers, B., Choksi, B., Bielawski, R., & VanRullen, R. (2021).** Does language help generalization in vision models? *ViGIL workshop, NAACL 2021*.
39. **Choksi, B., Mozafari, M., VanRullen, R., & Reddy, L. (2021).** Multimodal neural networks better explain multivoxel patterns in the hippocampus. In *Neural Information Processing Systems (NeurIPS) conference: 3rd Workshop on Shared Visual Representations in Human and Machine Intelligence (SVRHM 2021)*.
40. **Fakhe, C., VanRullen, R., Marque, P. & Dugué, L. (2021).** Causal link between the phase and amplitude of spontaneous alpha oscillations, cortical excitability and visual perception. *European Conference on Visual Perception (online)*.
41. **Chalvidal, M., Ricci, M., VanRullen, R., & Serre, T. (2021).** Go with the flow: Adaptive control for Neural ODEs. *ICLR 2021: International Conference on Learning Representations*.
42. **Choksi, B., Mozafari, M., Biggs O'May, C., Ador, B., Alamia, A. & VanRullen, R. (2020).** Brain-inspired predictive coding dynamics improve the robustness of deep neural networks. *NeurIPS 2020 Workshop SVRHM*
43. **Pang, Z., Choksi, B., Biggs O'May, C. & VanRullen, R. (2020).** Predictive coding results in perceived illusory contours in a recurrent neural network. *NeurIPS 2020 Workshop SVRHM*
44. **VanRullen, R. & Reddy, L. (2020).** Brain decoding with GANs. *WCNP: Winter Conference on Neural Plasticity*
45. **Mozafari, M., Reddy, L., & VanRullen, R. (2020).** Reconstructing Natural Scenes from fMRI Patterns using BigBiGAN. *Int. Joint Conf. Neural Networks (IJCNN)*
46. **Alamia, A., Gauducheau, V., Paisios, D. & VanRullen, R. (2019).** Which neural network architecture matches human behavior in artificial grammar learning? *Annual Conference on Cognitive Computational Neuroscience (CCN)*, Berlin (Germany).
47. **Schwenk, J., Zavitz, E., VanRullen, R., Price, N. S. & Bremmer, F. (2019).** Neural correlates of perceptual echoes in marmoset primary visual cortex. *Society for Neuroscience meeting (Chicago, USA)*.
48. **VanRullen, R. (2019).** Perceptual Cycles, Waves and Predictive Coding, Invited presentation, *Asia-Pacific Conference on Vision (Osaka, Japan)*.
49. **VanRullen, R. (2019).** Perceptual Cycles, Waves and Predictive Coding, Invited presentation, *York University VISTA - Centre for Vision Research International Conference on Predictive Vision, Toronto (Canada)*.
50. **VanRullen, R. (2019).** Perceptual Cycles, Waves and Predictive Coding, Invited presentation, *Dynamics of Vision and Touch (DyViTo)*, Rauschholzhäusen (Germany).
51. **Luo, C., Brüers, S., Berry, I., VanRullen, R. & Reddy, L. (2019).** fMRI signatures of perceptual echoes in early visual cortex. *Vision Sciences Society Annual Meeting, St Pete Beach (USA)*
52. **Merholz, G., VanRullen, R. & Dugué, L. (2019).** Oscillations modulate attentional search performance periodically. *Vision Sciences Society Annual Meeting, St Pete Beach (USA)*
53. **VanRullen, R. (2018).** Perceptual Cycles, Waves and Predictive Coding, Invited presentation, *Alpha Scales workshop, European Institute of Theoretical Neuroscience, Paris (France)*.
54. **Schwenk, J., VanRullen, R. & Bremmer, F. (2018).** The effects of short-term monocular deprivation on visual perceptual echoes. *Society for Neuroscience meeting (San Diego, USA)*.
55. **VanRullen, R. & Reddy, L. (2018).** Reconstructing faces from fMRI patterns using Generative Adversarial Networks. *Annual Conference on Cognitive Computational Neuroscience (CCN)*, Philadelphia (USA).
56. **Alamia, A. & VanRullen, R. (2018).** Predictive Coding Produces Alpha-band Rhythmic Travelling Waves. *Annual Conference on Cognitive Computational Neuroscience (CCN)*, Philadelphia (USA).
57. **VanRullen, R. (2018).** Perceptual Cycles, Waves and Predictive Coding, Invited presentation, *Symposium on New trends in decision-making: decision as inference? Paris (France)*.
58. **VanRullen, R. (2017).** Perceptual Cycles and Waves, Invited presentation, *Timing Research Forum, Strasbourg (France)*.
59. **Reddy, L., Cichy, R. & VanRullen, R. (2017).** Using DNNs as a yardstick for estimating the representational value of oscillatory brain signals. *Annual Conference on Cognitive Computational Neuroscience (CCN)*, New York (USA).
60. **VanRullen, R. (2017).** Predictive coding and neural communication delays produce alpha-band oscillatory Impulse Response Functions. *Annual Conference on Cognitive Computational Neuroscience (CCN)*, New York (USA).
61. **Reddy, L., Cichy, R. & VanRullen, R. (2017).** Oscillatory signatures of object recognition across cortical space and time. *40th European Conference on Visual Perception (ECPV)*, Berlin (Germany).
62. **Brüers, S. & VanRullen, R. (2017).** Does alpha power modulate perception independently of endogenous factors? *13th International Conference for Cognitive Neuroscience (ICON)*, Amsterdam (Netherlands).
63. **Reddy, L., Cichy, R. & VanRullen, R. (2017).** Oscillatory signatures of object recognition across cortical space and time. *Proceedings of the 17th Vision Sciences Society annual meeting, St Pete (Florida, USA)*.

64. **Brüers, S. & VanRullen, R. (2017).** At what latency does the phase of brain oscillations influence perception? Proceedings of the 17th Vision Sciences Society annual meeting, St Pete (Florida, USA).
65. **Gulbinaite, R., Roozendaal, D. & VanRullen, R. (2017).** Attention effects on steady-state visual evoked potentials in response to 3-80 Hz flicker. Proceedings of the 17th Vision Sciences Society annual meeting, St Pete (Florida, USA).
66. **Edwards, G., VanRullen, R. & Cavanagh, P. (2017).** EEG decoding of pre-saccadic effects on post-saccadic processing. Proceedings of the 17th Vision Sciences Society annual meeting, St Pete (Florida, USA).
67. **Benedetto, A., Lozano-Soldevilla, D. & VanRullen, R. (2017).** Ambient luminance changes modulate oscillatory properties of the visual system. Proceedings of the 17th Vision Sciences Society annual meeting, St Pete (Florida, USA).
68. **Chemla, S., Chavane, F. & VanRullen, R. (2016).** Revealing alpha oscillatory activity using voltage-sensitive dye imaging in monkey V1. European Conference on Visual Perception (Barcelona, Spain).
69. **Edwards, G., Marque, P., VanRullen, R. & Cavanagh, P. (2016).** Predictive position percepts mediated by parietal areas: TMS evidence. Proceedings of the 16th Vision Sciences Society annual meeting, St Pete (Florida, USA).
70. **Gulbinaite, R., Ilhan, B. & VanRullen, R. (2016).** Something out of nothing: The role of alpha-frequency reverberation in the triple-flash illusion. Proceedings of the 16th Vision Sciences Society annual meeting, St Pete (Florida, USA).
71. **Lozano-Soldevilla, D. & VanRullen, R. (2016).** The hidden spatial dimension of alpha: occipital EEG channels encode contralateral and ipsilateral visual space at distinct phases of the alpha cycle. Proceedings of the 16th Vision Sciences Society annual meeting, St Pete (Florida, USA).
72. **Crouzet, S. & VanRullen, R. (2016).** The half-time groove of divided attention: differences in EEG and decoding power spectra when attending to one vs. two items. Proceedings of the 16th Vision Sciences Society annual meeting, St Pete (Florida, USA).
73. **Bruers, S. & VanRullen, R. (2016).** Visual target detection in temporal white-noise: A “universal” forward model using oscillatory impulse response functions. Proceedings of the 16th Vision Sciences Society annual meeting, St Pete (Florida, USA).